

Flatcar Setup and Operations

Updates and Release Channels

Channels

Channel	Purpose
Alpha	Latest features; cutting edge
Beta	Release candidates; no known issues promoted from Alpha
Stable	Production-ready
LTS	18-month support; 12-month release cadence

Switch channels via Butane:

```
storage:
  files:
    - path: /etc/flatcar/update.conf
      contents:
        inline: |
          GROUP=beta
          SERVER=https://public.update.flatcar-linux.net/v1/update/
```

Update Flow

1. `update-engine.service` downloads update to the **inactive** USR partition
2. Reboot manager (`locksmithd` or `FLUO/kured` for Kubernetes) coordinates the reboot
3. GRUB boots into the updated partition
4. On failure, automatic rollback to previous partition

Update engine states: `IDLE` → `CHECKING_FOR_UPDATE` → `UPDATE_AVAILABLE` → `DOWNLOADING` → `VERIFYING` → `FINALIZING` → `UPDATED_NEED_REBOOT`

Reboot Strategies (`locksmithd`)

Strategy	Behavior
<code>reboot</code> (default)	Reboot 5 minutes after update is applied
<code>etcd-lock</code>	Acquire distributed etcd lock before rebooting

Strategy	Behavior
off	Never reboot automatically

For Kubernetes: Use [FLUO](#) or [kured](#) instead of locksmithd.

Verifying Images

Flatcar images are GPG-signed. Verify before use:

```
gpg --verify flatcar_production_image.vmlinuz.sig
```

Clusters

- **Discovery:** etcd-based cluster discovery using a discovery URL
- **Architectures:** single-master, multi-master (HA), or worker-only node roles
- **ECS:** Flatcar nodes can register with Amazon ECS via the `booting-on-ecs` guide

Customization

Network (networkd)

Flatcar uses `systemd-networkd` for network configuration. Custom network configs go in `/etc/systemd/network/`. Example via Butane:

```
storage:
  files:
    - path: /etc/systemd/network/10-static.network
      contents:
        inline: |
          [Match]
          Name=eth0
          [Network]
          Address=192.168.1.100/24
          Gateway=192.168.1.1
```

systemd

Topic	How
Drop-in units	Files in <code>/etc/systemd/system/NAME.d/</code>
Environment variables	<code>EnvironmentFile=</code> or <code>Environment=</code> in unit

Topic	How
Timers	<code>.timer</code> units as cron replacements
udev rules	Files in <code>/etc/udev/rules.d/</code>
User units	<code>~/.config/systemd/user/</code>

Other Customization

- **Users:** Defined in Butane `passwd.users` ; default user is `core` (sudo via wheel group)
- **DNS:** `/etc/systemd/resolved.conf` or networkd `.network` file `[Network]DNS=`
- **Date/timezone:** `timedatectl set-timezone` ; NTP via `systemd-timesyncd`
- **ACPI:** Custom ACPI tables via `/etc/acpi/`
- **NVIDIA GPUs:** Via the `nvidia-drivers-*` sysex (available since 4344.0.0)
- **Power management:** Tuned profiles or systemd power targets
- **OverlayBD:** Artifact streaming extension (available since 4426.0.0)

Security

Built-in Hardening

- `/usr` is read-only and dm-verity protected
- No package manager — reduced attack surface
- Only `sshd` (port 22) listens by default
- Single default user `core` (wheel group, sudo access)

Key Security Features

Feature	Notes
LUKS disk encryption	Full-disk encryption via <code>luks</code> setup in Ignition
SELinux	Supported; Flannel >0.17.0 has known incompatibility
FIPS	Available in LTS edition
SSSD	Centralized auth (AD, LDAP)
SSH hardening	Custom sshd config via <code>/etc/ssh/sshd_config.d/</code>
Certificate authorities	Add custom CAs to <code>/etc/ssl/certs/</code>
SMT disable	<code>nosmt</code> kernel parameter
Audit subsystem	<code>auditd</code> available
Self-signed certs	<code>cfssl</code> tooling

Feature	Notes
Trusted hardware	TPM requirements documented

Hardening the `core` user (require password for sudo):

```
storage:
  files:
    - path: /etc/sudoers.d/core-passwd
      mode: 0644
      contents:
        inline: |
          core    ALL=(ALL)    ALL
```

(With no password set on `core` , sudo is effectively blocked.)

Storage

Topic	Notes
RAID	Software RAID via <code>mdadm</code> ; recommended for root filesystem
Adding disk space	Partition and format via Ignition or runtime <code>fdisk / mkfs</code>
Mounting storage	Systemd <code>.mount</code> units or <code>/etc/fstab</code> entries
iSCSI	<code>open-iscsi</code> included
ZFS	Available as opt-in <code>sysex</code> (<code>zfs</code>) since 3913.0.0
LVM	<code>lvm2</code> tools included
Swap	Swap file or swap partition via Ignition
btrfs	Default root filesystem format

Debugging

Tool/Method	Notes
Install debugging tools	<code>toolbox</code> container for accessing host tools not in base image
System log	<code>journalctl -u SERVICE</code> or <code>journalctl -b</code>
Crash logs	<code>/var/log/crash/</code>
btrfs troubleshooting	<code>btrfs check</code> , <code>btrfs scrub</code>
Manual rollback	Edit GRUB to boot inactive partition; <code>cgpt prioritize</code>

toolbox drops you into a Fedora container with full host filesystem access:

```
toolbox
# now inside container with /media/root as host filesystem
```

Manual rollback:

```
# List partitions and priorities
cgpt show /dev/sda
# Prioritize the inactive USR partition
cgpt prioritize -P 1 -i 4 /dev/sda
```